



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System

TAILWIND GUIDE

Project	Smart Medicine Availability & Intelligent Janaushadhi Recommendation System
Document	TAILWIND GUIDE
Version	v1.0.0
Date	02 July 2026

How Tailwind is Used in This Project

1. Why Tailwind CSS?

Traditional CSS approach:

```
/* styles.css */
.medicine-card { background: white; border-radius: 12px; padding: 16px; }
.medicine-card:hover { box-shadow: 0 8px 24px rgba(0,0,0,0.12); }
```

Problem: You have two files to maintain. CSS files grow huge. Class names can conflict.

Tailwind approach:

```
<div className="bg-white rounded-xl p-4 hover:shadow-card transition-shadow">
```

All styling is inline in the JSX. No context-switching between files. No class name conflicts.

Why Tailwind was chosen for this project:

1. **Speed** — build UIs very quickly using utility classes
 2. **Consistency** — the design system is enforced by the token system
 3. **No CSS files** — no separate stylesheets to maintain
 4. **Responsive built-in** — `sm:`, `md:`, `lg:` prefixes
 5. **Design system** — `@theme {}` in `index.css` defines all tokens
-

2. How Tailwind v4 Works (CSS-First Design System)

This project uses **Tailwind v4** with a **CSS-first design system**. The custom design tokens are defined in `src/index.css`:

```
@import "tailwindcss";

@theme {
  /* Colors */
  --color-primary-600: #2563eb;
  --color-success-500: #22c55e;

  /* Typography */
  --font-sans: 'Inter', ui-sans-serif;

  /* Spacing */
  --spacing-section-y: 5rem;
}
```

These tokens become Tailwind utility classes automatically:

- `--color-primary-600` → `text-primary-600`, `bg-primary-600`, `border-primary-600`
 - `--color-success-500` → `text-success-500`, `bg-success-500`
-

3. Flexbox

Flexbox is the most-used layout system in this project.

Common patterns:

```

{/* Center content horizontally and vertically */}
<div className="flex items-center justify-center h-64">
  <Spinner />
</div>

{/* Space items with gap */}
<div className="flex items-center gap-3">
  <Avatar />
  <span>{user.name}</span>
</div>

{/* Push items to opposite ends */}
<div className="flex items-center justify-between">
  <h1>Dashboard</h1>
  <Button>Add New</Button>
</div>

{/* Wrap badges on overflow */}
<div className="flex flex-wrap gap-1.5">
  <Badge>In Stock</Badge>
  <Badge>Generic</Badge>
  <Badge>Jan Aushadhi</Badge>
</div>

```

Class	Effect
flex	Enable flexbox
flex-col	Stack children vertically
flex-row	Arrange children horizontally
items-center	Align on cross axis (vertically in row)
justify-center	Align on main axis (horizontally in row)
justify-between	Space items to opposite ends
flex-1	Grow to fill available space
flex-wrap	Allow wrapping to next line
gap-3	Spacing between children (12px)
shrink-0	Prevent item from shrinking

4. Grid

Used for card grids and multi-column layouts.

```

{/* Responsive card grid */}
<div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-4">
  {medicines.map(m => <SearchResultCard key={m.id} medicine={m} />)}
</div>

{/* Two-column form */}
<div className="grid grid-cols-1 md:grid-cols-2 gap-6">
  <Input label="First Name" />

```

```
<Input label="Last Name" />
</div>
```

5. Spacing

Tailwind uses a spacing scale based on 0.25rem units:

- `p-4` = `padding: 1rem` (4 × 0.25rem)
- `m-2` = `margin: 0.5rem`
- `gap-3` = `gap: 0.75rem`

Spacing in this project:

```
{/* Card padding */}
<div className="p-4">           {/* 16px padding all sides */}
<div className="px-4 py-3">      {/* 16px horizontal, 12px vertical */}
<div className="p-6">           {/* 24px – larger cards */}

{/* Section spacing */}
<section className="py-20">      {/* 80px vertical padding for sections */}

{/* Small gap */}
<div className="gap-1.5">        {/* 6px between badges */}
<div className="gap-4">          {/* 16px between cards */}
```

6. Typography

```
{/* Page title */}
<h1 className="text-3xl font-bold text-slate-900">Medicine Search</h1>

{/* Section subtitle */}
<p className="text-lg text-slate-600">Find affordable medicines</p>

{/* Card name */}
<h3 className="text-sm font-bold text-slate-900 truncate">{medicine.name}</h3>

{/* Muted helper text */}
<p className="text-xs text-slate-400">{medicine.composition}</p>

{/* Price */}
<span className="text-base font-extrabold text-slate-900">■{price}</span>

{/* Strikethrough MRP */}
<span className="text-xs text-slate-400 line-through">■{mrp}</span>
```

Class	Size
text-xs	12px
text-sm	14px
text-base	16px
text-lg	18px
text-xl	20px
text-2xl	24px

Class	Size
text-3xl	30px

7. Colors

This project uses a **Medical Blue + Teal** colour palette defined in [index.css](#):

```

{/* Primary medical blue */}
<button className="bg-primary-600 hover:bg-primary-700 text-white">
  View Details
</button>

{/* Success green */}
<Badge className="bg-success-100 text-success-700">In Stock</Badge>

{/* Warning amber */}
<Badge className="bg-warning-100 text-warning-700">Limited Stock</Badge>

{/* Danger red */}
<Badge className="bg-danger-100 text-danger-700">Out of Stock</Badge>

{/* Neutral slate */}
<p className="text-slate-500">Secondary text</p>
<p className="text-slate-400">Muted text</p>

```

8. Responsive Classes

Tailwind's mobile-first responsive system uses breakpoint prefixes:

Prefix	Min-width	Device
(none)	0px	Mobile
sm:	640px	Large mobile
md:	768px	Tablet
lg:	1024px	Laptop
xl:	1280px	Desktop
2xl:	1536px	Large desktop

Example from SearchResultsPage:

```

{/* 1 column on mobile, 2 on tablet, 3 on desktop, 4 on large desktop */}
<div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-4">

```

Example from Sidebar:

```

{/* Hidden on mobile, shown on large screens */}
<aside className="hidden lg:flex w-64 flex-col">

```

9. Hover and Focus States

Every interactive element has hover and focus styles for UX and accessibility:

```

{/* Button hover */}
<button className="bg-primary-600 hover:bg-primary-700 transition-colors">

{/* Card hover */}
<div className="shadow-sm hover:shadow-md hover:-translate-y-0.5 transition-all duration-200">

{/* Link hover */}
<a className="text-slate-400 hover:text-primary-600 transition-colors">

{/* Focus ring for keyboard users */}
<button className="focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-primary-500">

```

focus-visible: only shows the ring when using keyboard — not on mouse click (cleaner UX).

10. Transitions and Animations

```

{/* Smooth color change */}
<button className="transition-colors duration-200">

{/* Card lift on hover */}
<div className="transition-all duration-200 hover:-translate-y-0.5 hover:shadow-md">

{/* Page enter animation (applied to all layouts) */}
<main className="page-enter">

```

The **page-enter** animation is defined in **index.css**:

```

@keyframes page-enter {
  from { opacity: 0; transform: translateY(8px); }
  to   { opacity: 1; transform: translateY(0); }
}
.page-enter {
  animation: page-enter 200ms ease-out;
}

```

11. Cards

The standard card pattern used throughout:

```

<div className="
  bg-white
  rounded-xl
  border border-slate-100
  shadow-sm
  hover:shadow-md
  hover:-translate-y-0.5
  transition-all duration-200
  p-4
">
  {/* Card content */}
</div>

```

Premium / highlighted card (Best Value):

```
<div className="border-primary-400 ring-2 ring-primary-200">
```

12. Forms Styling

```

{/* Input field */}
<input className="
  w-full
  px-3 py-2
  text-sm
  border border-slate-200
  rounded-lg
  focus:outline-none
  focus:ring-2
  focus:ring-primary-500
  focus:border-transparent
  transition-shadow
" />

{/* Error state */}
<input className="border-danger-400 focus:ring-danger-400" />

{/* Label */}
<label className="block text-sm font-medium text-slate-700 mb-1">

```

13. Dark Mode

The theme system applies `dark` class to `<html>` via ThemeContext. Tailwind's `dark:` prefix activates dark variants:

```
<div className="bg-white dark:bg-slate-900 text-slate-900 dark:text-slate-100">
```

The design token system in `index.css` also overrides `@theme` values under `.dark {}` for seamless theming.

14. Utility Combinations (Real Examples)

Skip link (index.html):

```

.skip-link {
  position: absolute;
  top: -100%; /* Hidden by default */
}
.skip-link:focus {
  top: 0; /* Visible when focused by keyboard */
}

```

Badge with dot:

```

<span className="inline-flex items-center gap-1.5 px-2 py-0.5 rounded-full text-xs font-medium bg-success-100 t
  <span className="w-1.5 h-1.5 rounded-full bg-success-500" />
  In Stock
</span>

```

Truncated text:

```
<h3 className="text-sm font-bold text-slate-900 truncate max-w-full">
  {medicineName}
</h3>
```